🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.6

Represent Rational Numbers and Their Opposites on the Number Line

Lesson 7-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can place rational numbers, including fractions and decimals, on a number line.
Language Objective: I can explain my reasoning using the words rational number, fraction, decimal, and number line.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Rational number | Fraction | Decimal | Number line | Equivalent | Integer

👆 *Tap any word to see its meaning and a picture.*

1. A number you can write as a fraction of two integers, with a bottom number that is not zero, is a ______________.

2. A number that shows part of a whole, like 3/5, is a ______________.

3. A number that uses a point to show parts of a whole is a ______________.

4. A line with numbers in order used to place and compare values is a ______________.

5. Numbers that name the same value, like 1/2 and 0.5, are ______________.

6 A whole number or its opposite is an ______________ .

Write About the Math

Put the four readings in order from lowest water to highest. How do the decimals let the keeper say something a whole number could not?

Pon las cuatro lecturas en orden del agua más baja a la más alta. ¿Cómo le permiten los decimales al encargado decir algo que un número entero no podría?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Rational number** *Número racional*
- ☐ **Fraction** *Fracción*
- ☐ **Decimal** *Decimal*
- ☐ **Number line** *Recta numérica*
- ☐ **Equivalent** *Equivalente*

Model: *-1 1/2 falls between ____ and ____ because ____.* — Student places -1 1/2 between -1 and -2, halfway, reasoning from the value of the fraction part.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The lowest reading was ____ and the highest was ____. Tuesday's ____ ft means the water sat $1\frac{1}{4}$ feet ____ the mark, which is between −1 and ____ on the number line.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Rational number
2. **Notes 2:** Fraction
3. **Notes 3:** Decimal
4. **Notes 4:** Number line
5. **Notes 5:** Equivalent
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Rational Numbers on the Number Line is treating the numerator and denominator of a negative fraction as two separate whole numbers instead of one value between two integers — for example, plotting -3/4 by going to -3 and then counting 4 more units to the left, landing (incorrectly) at -7, instead of recognizing that -3/4 = -0.75, which sits between 0 and -1. Before you submit an answer, find the two whole numbers your value falls between, then split that space into equal parts.*
8. **Try It 1:** A. -1.5, -0.5, 0.5 — *The farthest left on the number line comes first: -1.5, then -0.5, then 0.5.*
9. **Try It 2:** C. -1/4 — *-3/4 = -0.75 and -1/4 = -0.25. Since -0.75 is farther left (less), -3/4 is to the left of -1/4.*